



EMOTION CLASSIFICATION

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11.04.2025



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Problem

Understanding the Emotion Layer

from media content (movies, TV shows, video clips)

The **Content Intelligence Agency (CIA)** specializes in building tools to **analyze content** using AI tools

Develop a reliable tool for **emotion classification**



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Solution

Pipeline to analyze:

Core Emotions: **7**

Fine-Grained Emotions: **19**

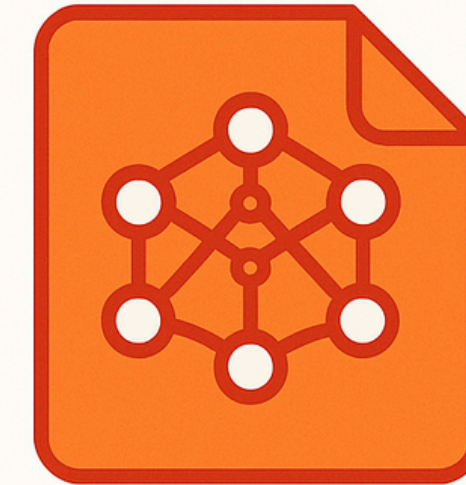
Emotion Intensity

Why our solution?

Scalable on any language

Provides **nuanced** emotions

Outperforms LLMs on emotion analysis



**EMOTION
CLASSIFICATION
USING
TRANSFORMERS
& MACHINE LEARNING**

NLP

 **HAPPY**
 **SAD**
 **SURPRISE**
 **ANGRY**



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Pipeline

High-Level Flow:

User input: **YouTub**e link or **MP3** file path

Extract audio

Transcribe audio (**Whisper**)

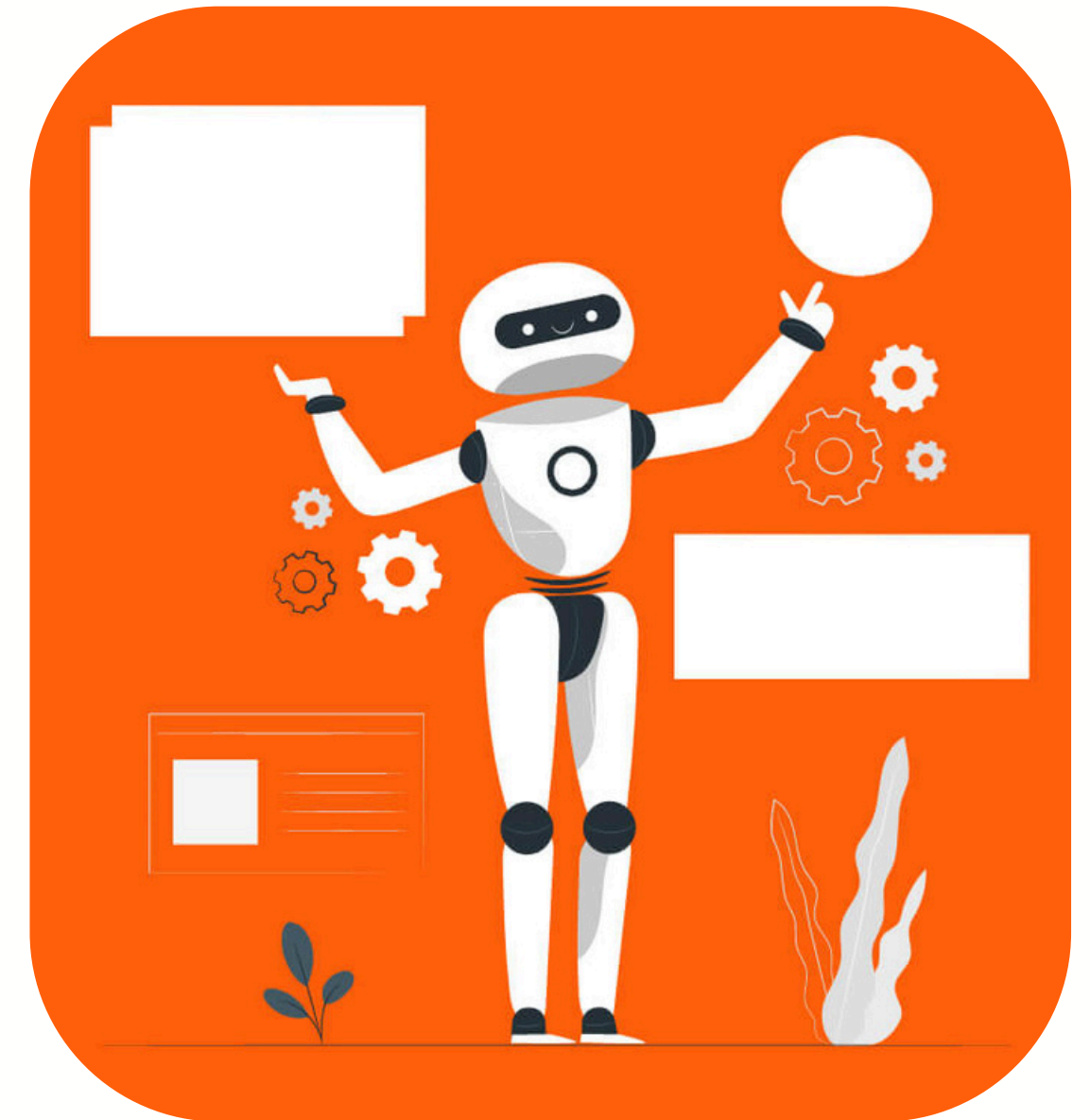
Correct transcriptions (**Llama 3.3**)

Translate transcription to English

Calculate **sentiment intensity**

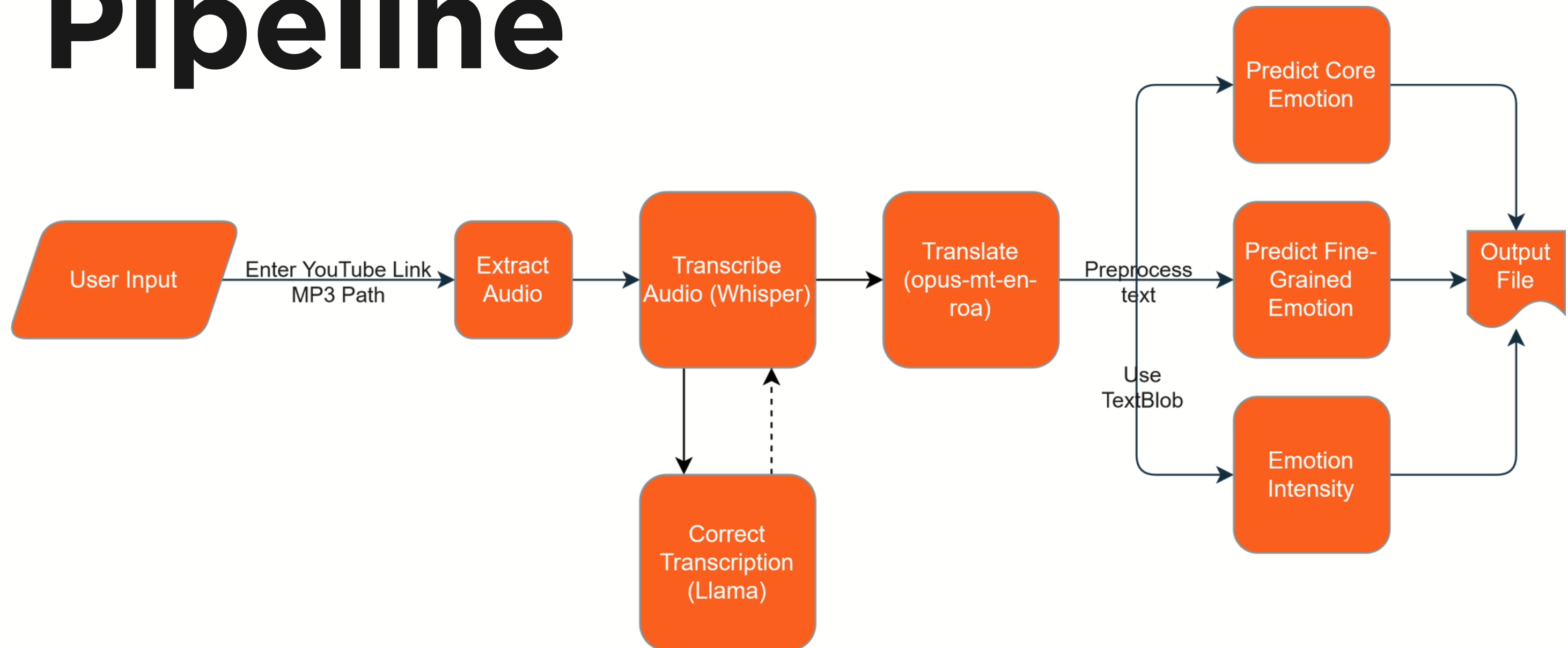
Predict core and **fine-grained** emotion

Generate **output** file





Pipeline





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Model Performance

Transformers

BERT, distilBERT, **RoBERTa**, **ALBERTo**

Best model: **RoBERTa**

F1-Score (weighted): **70%**

Accuracy: **69%**

High Precision: ("**happiness**", "**neutral**")

Low Precision: ("**disgust**", "**fear**")

Learning Rate: **3e-6** (**0.000003**)

Batch Size: **16**

Epochs: **10**

Strengths: Captures **context** effectively using **attention mechanism**

Weaknesses: Requires **high** computational resources



Model Performance

Other models

Logistic Regression, Naive Bayes, LSTM, RNN

Best model: Logistic Regression

Hyperparameters: max_iter=1000, RandomizedSearchCV,
Best Parameters: {'C': 1.49, 'penalty': 'l1', 'solver': 'liblinear'}

F1-Score: **57%**

Strengths: **low cost**, interpretability

Weakness: low accuracy, struggles with **nuanced emotions**



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Ethics

XAI findings:

Token-Level Focus Over Phrase Meaning
Modifiers Ignored
Overreliance on **Punctuation/Special Tokens**
False Confidence in **Noisy Inputs**
Tokenization **Distorts Emotion**
Emotion **Word** Misjudgment

Environmental Impact:

- CO₂: ~0.049 kg/session; ~0.3 kg/year (1K daily uses)
- Suggestion: early stopping, FP16, use TinyBERT for inference



Results

Performance Metrics and Comparative Gains

Three models:

Translation model Helsinki-NLP/opus-mt-roa-en

Core emotion classifier (7 Emotions): F1-Score = ~**70%** (R.G.: ~**14.3%** => **X4.9 better**)

Fine-Grained classifier (19 emotions): F1-Score = ~**32.6%** (R.G.: ~**5.26%** => **X6.2 better**)

Sentence	Corrected_Sentence	Translation	Preprocessed_Translation	Core_Emotion	Fine_Emotion	Intensity
Vă iubesc mult!	Vă iubesc mult!	I love you so much!	i love you so much!	happiness	peaceful	moderate
O să câștigăm, o să câștigăm.	O să câștigăm, o să câștigăm.	We're gonna win, we're gonna win.	we are going to win, we are going to win	happiness	optimistic	very intense
Și mâncăm, și mâncăm.	Și mâncăm, și bem.	And we eat, and we drink.	and we eat, and we drink	neutral	neutral	low
Și mâncăm, și mâncăm.	Și mâncăm, dar mereu tot mâncăm.	And we eat, but we always eat.	and we eat, but we always eat	neutral	neutral	low



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Limitations

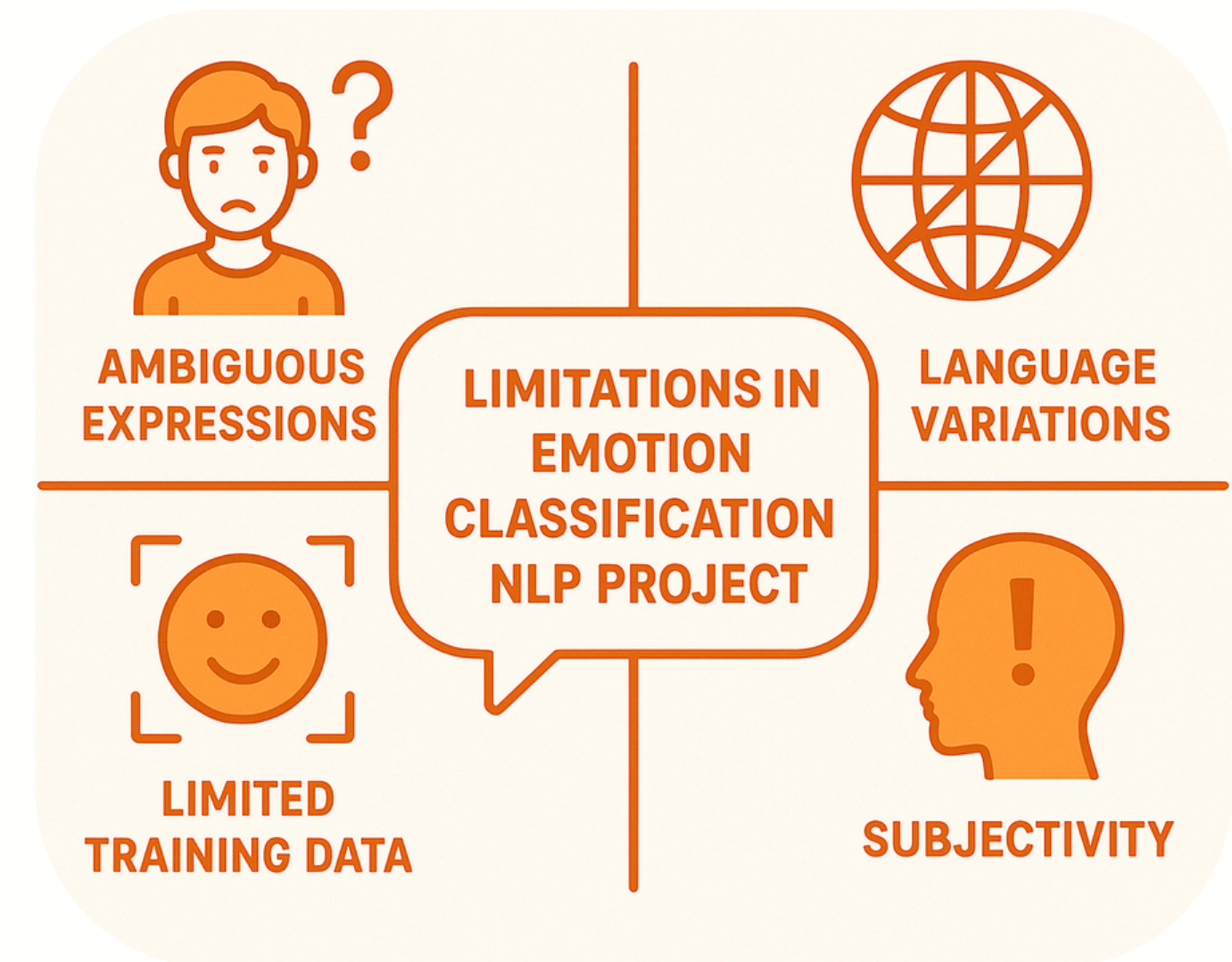
Emotion Classification

Natural Language -> ambiguous, nuanced, contextual

Words -> multiple meanings, connotations, cultural interpretations

Emotions -> tone, irony, idioms, and context.

The same phrase can evoke **different emotions** depending on delivery.





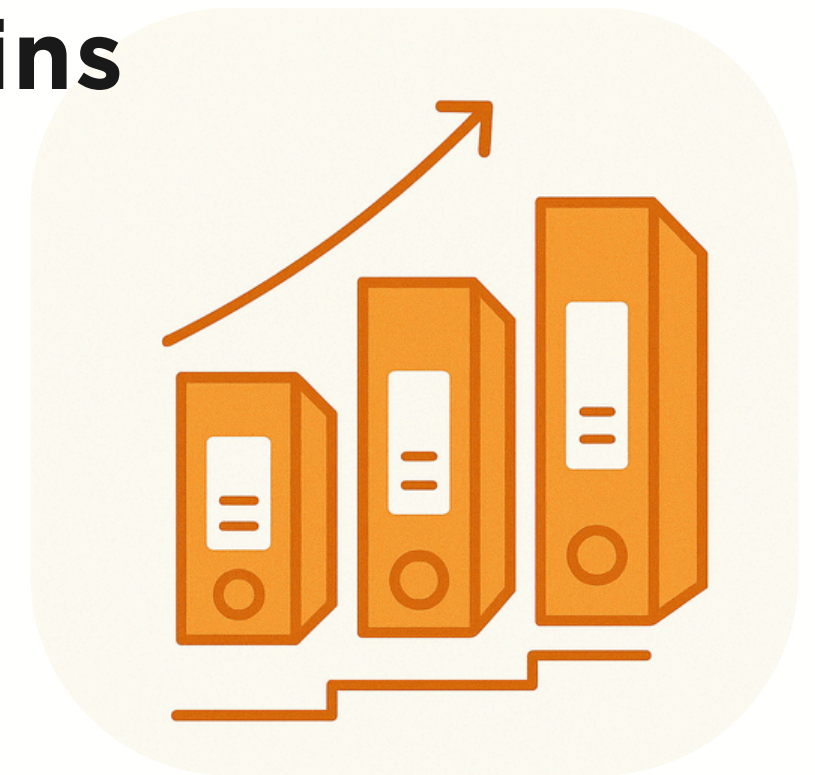
Recommendations

Performance Metrics and Comparative Gains

Language variety dialects, slang, informal language

Multilingual support

Multi-Modal Emotion Prediction - **Audio, Video, Text**





QUESTIONS?

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**THANK
YOU**

